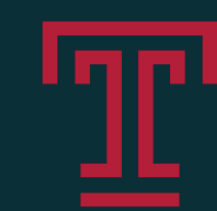


CypherLens

An Interactive Demo for Evaluating and Diagnosing NL-to-Cypher Systems



Zahra Mohammadi · Ali Ansari · Longin Latecki · Eduard Dragut

Temple University, Philadelphia, PA, USA · PVLDB 19(12) · VLDB 2026, Boston





Code, data & demo

github.com/YasmeenMohammadi/CypherLens

Why execution accuracy falls short

C1

Metric brittleness

Strict result equality punishes queries that differ only in aliases, column order or variable names.
Gold returns 1937 under min(n.inception_year); the model returns 1937 as earliest_founding_year → EA = 0.

C2

No result normalization

Cypher returns nested, heterogeneous JSON, not flat tables.
n.name, n.gender vs. p.name AS founder, p.gender AS gender hold identical data yet look mismatched; nulls, timeouts and empty results are unhandled.

C3

Error opacity & query blindness

A 0/1 signal never says why a query failed: wrong string literal? missing OPTIONAL MATCH? reversed traversal? Aggregate scores also hide whether failures sit in retrieval, aggregation or multi-hop reasoning.

C4

Binary scoring

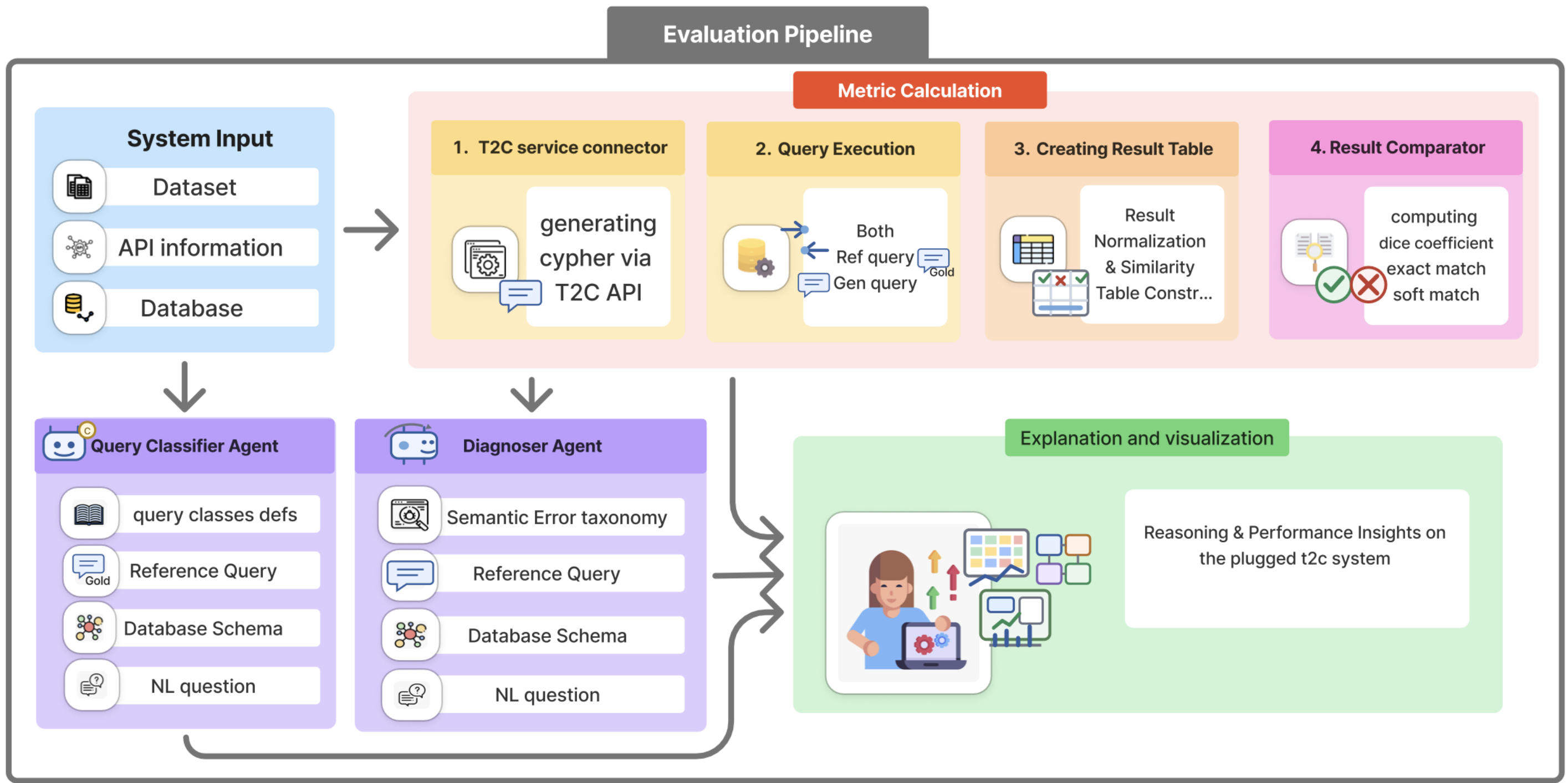
A near-miss scores the same as nonsense. Asked for Czech-HQ CEOs, a query that returns the right CEOs plus company, start and end year still gets 0.

C5

Coincidental correctness

A structurally wrong query can return the right rows on this dataset – e.g. ignoring a player's Lakers tenure when filtering awards – and EA calls it correct, hiding a bug that surfaces when data changes.

System overview: a black-box evaluation pipeline



Inputs: NL questions with gold Cypher · Neo4j connection · API access to the target T2C system (no access to model internals). **Pipeline:** ① generate Cypher via the T2C API → ② execute generated and gold queries → ③ normalize results and build a Similarity Table → ④ compute Exact Match, Soft Match and Dice. Classifier and Diagnoser Agents run through a provider-agnostic LLM routing layer.

Inside CypherLens

1

Execution & schema-aware evaluation

Retrieves node labels, relationship types, property keys and directions from Neo4j, then aligns generated and gold projections through staged matching: exact columns → alias normalization → type-aware → schema-aware, with embedding alignment only as a fallback (e.g. sum(x)/size(collect(x)) ↔ avg(x)). The Result Comparator turns the table into Exact Match, Soft Match and Dice.

2

Query Type Classifier Agent

Assigns every instance to a canonical class – simple retrieval, aggregation, multi-hop traversal, path query – from the question, schema and gold query. Reuses dataset labels when they exist, so annotated and unannotated benchmarks both get per-class metric and error breakdowns.

3

Diagnoser Agent

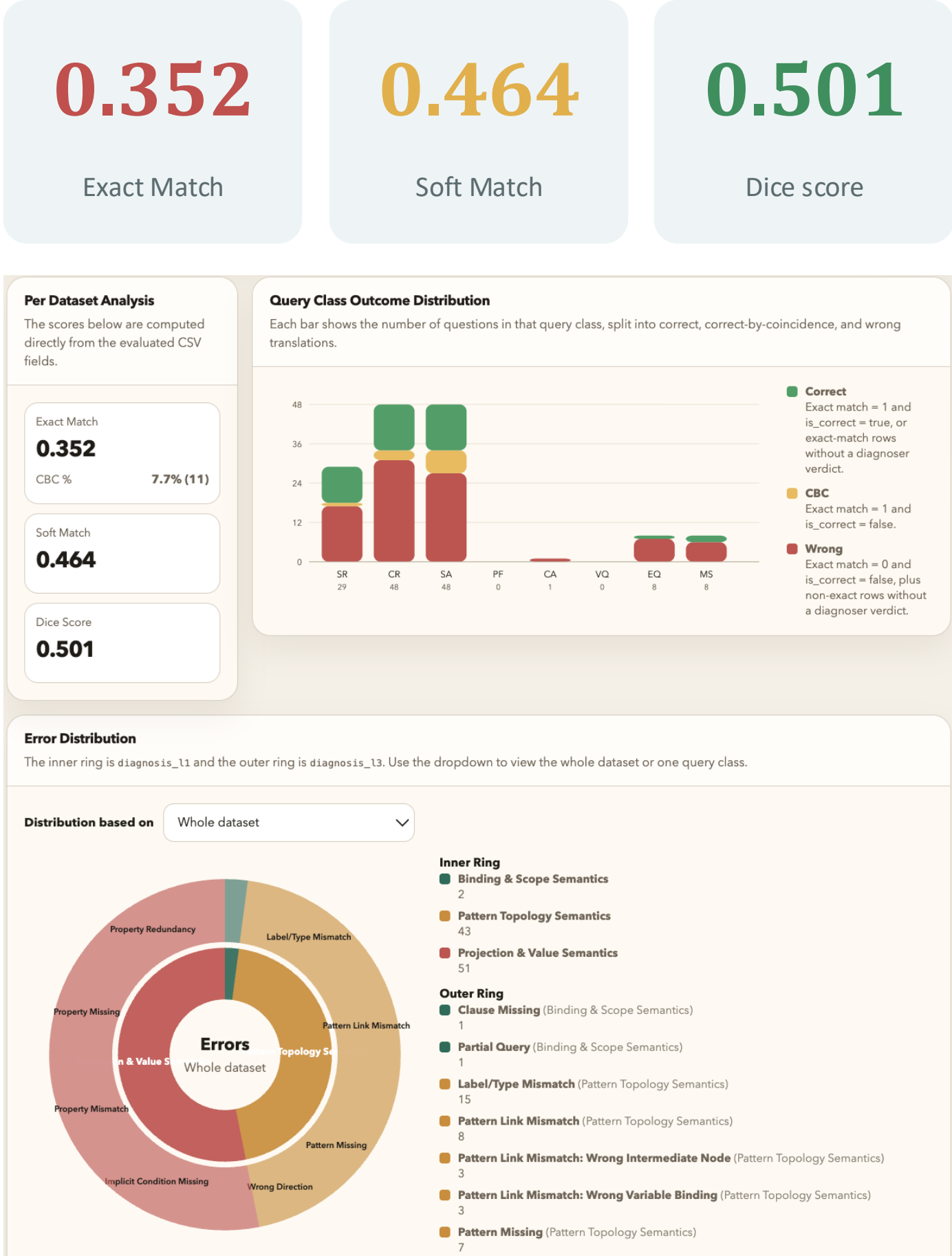
Invoked whenever EM = 0. Reasons over question, schema, gold and generated queries plus sampled results, then emits a label from a hierarchical Cypher error taxonomy (projection & value, pattern topology, binding & scope, clause-level – e.g. Direction Mismatch, Wrong Hop Count, Implicit Condition Missing) with a confidence score and grounded rationale. Also audits EM = 1 cases for coincidental correctness.

4

Interactive visualization interface

Dataset level: aggregate metrics, per-class outcome bars and hierarchical error rings. Query level: question, gold and generated Cypher side by side with execution outputs, the similarity table, metric scores, class label and the diagnosis report.

What the dashboard reveals



Read the gap: Soft Match 0.464 vs. Exact Match 0.352 means many “failures” are near-misses, not wrong answers.
Where errors live: Projection & Value (51) and Pattern Topology (43) semantics dominate; label/type mismatch, pattern-link mismatch and missing patterns concentrate in multi-hop classes.
Hidden bugs: 11 EM = 1 queries (7.7%) are CBC candidates.

Demonstration scenarios

1

Setup & dataset-level evaluation

Point CypherLens at a dataset, a database and a T2C API; toggle the Diagnoser, Classifier and CBC inspection and pick an LLM backend. One click runs the full pipeline with streamed progress and logs, then opens the dashboard: graded scores, per-class outcomes and error distributions (e.g. simple retrieval solved, pattern-topology errors dominate multi-hop).

2

Per-query debugging

Open any instance to see the question, gold and generated Cypher side by side with their results and the similarity table. The Diagnoser explains, for example, a one-hop relationship where a two-hop traversal was required (Wrong Hop Count); Soft Match 0.5 shows the core result is partially right where EA gives 0.

3

Correct-by-coincidence inspection

Filter to EM = 1 ∧ CBC-flagged. A generated query links employees straight to cities, skipping the intermediate project node the gold path requires – identical results on this data, but a Pattern Topology error the Diagnoser catches. About 7% of EM = 1 queries are flagged.